**MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY**

**PHYS-111 Lecture 3 Tutorial questions**

1. An airplane flies 250 km due west from city A to city B and then 420 km in the direction of 60 degrees north of west from city B to city C.

   (a) In straight-line distance, how far is city C from city A?

**Step 1**

1 of 2

$\mathbf{(a)}$ Here the directions are shown in the figure below and let +ve x-axis is along west and +ve y-axis is along north.

Here $\vec{\mathrm{P}} = 2.00 \times 10^2\,\mathrm{km}\,\hat{\mathrm{i}}$.

$\vec{\mathrm{Q}} = 3.00 \times 10^2\,\mathrm{km}\,\hat{\mathrm{i}} + 3.00 \times 10^2\,\mathrm{km}\,\hat{\mathrm{j}}$.

Now from figure

$$\begin{aligned} \mathrm{AO} &= \mathrm{AB} + \mathrm{BO} \\ &= 2.00 \times 10^2\,\mathrm{km} + 3.00 \times 10^2\,\mathrm{km} \times \cos 30^\circ \\ &= 459.81\,\mathrm{km} \\ &= 4.60 \times 10^2\,\mathrm{km} \end{aligned} \tag{1}$$

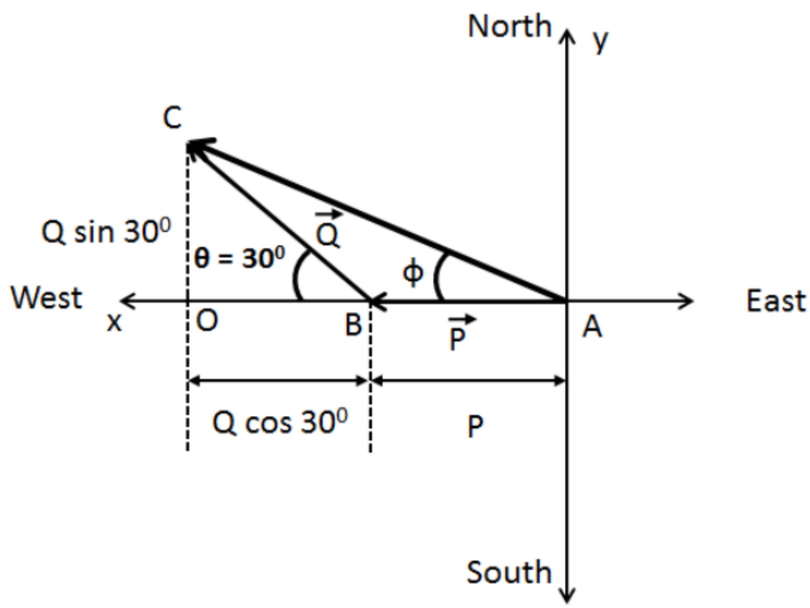
And

$$\begin{aligned} \mathrm{OC} &= Q \sin 30^\circ \\ &= 3.00 \times 10^2\,\mathrm{km} \times \sin 30^\circ \\ &= 150\,\mathrm{km} \\ &= 1.50 \times 10^2\,\mathrm{km} \end{aligned} \tag{2}$$

Therefore the straight-line distance between city A and city C is

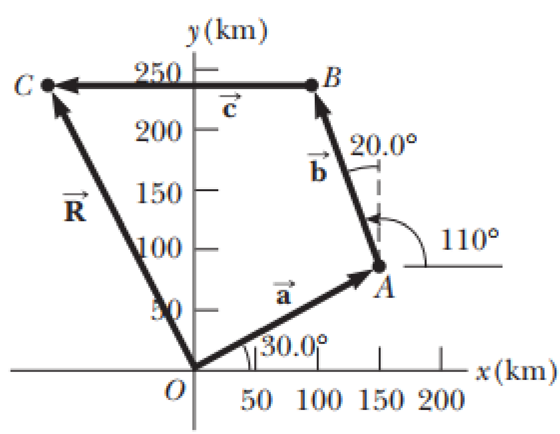
$$\begin{aligned} AC &= \sqrt{(AO)^2 + (OC)^2} \\ &= \sqrt{(4.60 \times 10^2 \text{ km})^2 + (1.50 \times 10^2 \text{ km})^2} \\ &= 4.838 \times 10^2 \text{ km} \\ &\approx 4.84 \times 10^2 \text{ km} \end{aligned}$$

Here significant number is three.




(b) Relative to city A, in what direction is city C?

2. A commuter airplane starts from an airport and takes the route shown in Figure below. The plane first flies to city A, located 175 km away in a direction 30.0 north of east. Next, it flies for 150 km 20.0 west of north, to city B. Finally, the plane flies 190 km due west, to city C. Find the location of city C relative to the location of the starting point.



$a_x$ = $a\cos\theta_x$ = 175km cos30° = 175km 0.866 = 152 km

$a_y$ = $a\sin\theta_y$ = 175km sin30° = 175km 0.500 = 87.5 km

$b_x$ = $b\sin\theta_x$ = -150km sin20° = -150km 0.342 = -51.3 km

$b_y$ = $b\cos\theta_y$ = 150km cos20° = 150km 0.940 = 141 km

$c_x$ = $c\cos\theta_x$ = 190km cos180° = 190km (-1) = - 190 km

$c_y$ = $c\sin\theta_y$ = 190km sin180° = 0

$R_x$ = $a_x$ + $b_x$ + $c_x$ = 152 - 51.3 – 190 km = -89.3. km

$R_y$ = $a_y$ + $b_y$ + $c_y$ = 87.5 + 141 + 0 = 229. km.

R = sqrt($R_x^2$ + $R_y^2$) = sqrt(89.3² +229²) = sqrt(7974.5+52441)=sqrt(60415) 246 km.

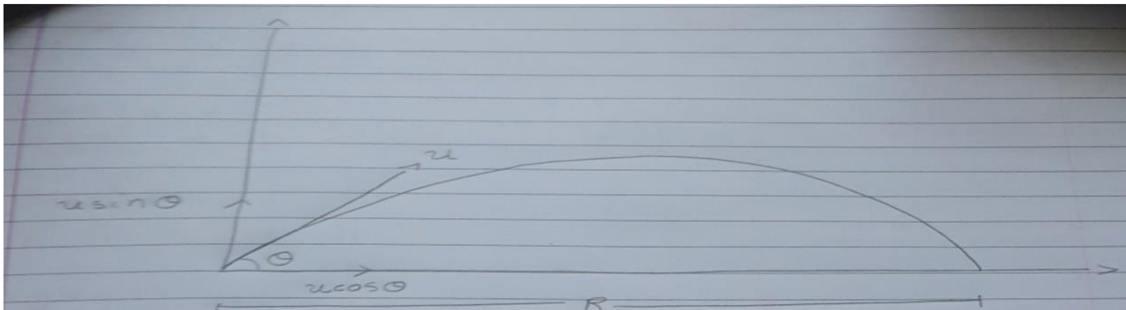
$\theta_R$ = $\tan^{-1}$ ($R_y/R_x$) = $\tan^{-1}$ (229km/-89.3km) = $\tan^{-1}$ (-2.564) = 90° -39° = 51° to –x axis.

3. Show that the maximum height of a projectile is given by $y_{max} = \frac{V_i^2 Sin^2\theta_i}{2g}$

4. A projectile is fired from the ground with an initial velocity $V_i$ initial angle $\theta_i$ lands on the same level as it was fired. Show that;

   a. Show that the range **R** of the projectile is $R = \frac{V_i^2 Sin\ 2\theta_i}{g}$

   b. Determine the maximum range $R_{max}$ of the projectile and the angle at which the maximum range occurs.




$R = u\cos\theta \times t$

$t = \frac{u\sin\theta}{g}$

$\therefore\ R = \frac{u^2 \sin\theta\cos\theta}{g}$

$R = \frac{u^2 \sin 2\theta}{2g}$

For R to be maximum sin2θ should be maximum

sin 90° = 1 is max value for sine.

$\therefore\ \sin 2\theta = \sin 90°$

$\therefore\ \boxed{\theta = 45°}$

5. A projectile is fired in such a way that its horizontal range is equal to three times its maximum height. What is the angle of projection? Give your answer to three significant figures.

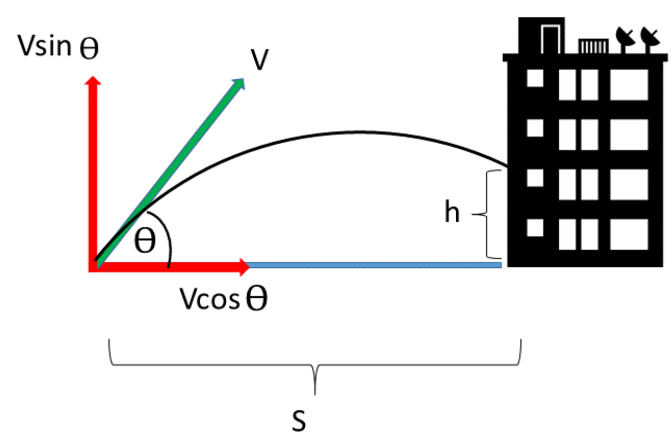
$$h = \frac{v_i^2 \sin^2 \theta_i}{2g};\, R = \frac{v_i^2(\sin 2\theta)}{g};\, 3h = R$$

$$\text{so, } \frac{3v_i^2 \sin^2 \theta_i}{2g} = \frac{v_i^2(\sin 2\theta_i)}{g}$$

$$\text{or, } \frac{2}{3} = \frac{\sin^2 \theta_i}{\sin 2\theta_i} = \frac{\tan \theta_i}{2}$$

$$\text{thus, } \theta_i = \tan^{-1}\left(\frac{4}{3}\right) = 53.1^0$$

6. A firefighter 50.0 m away from a burning building directs a stream of water from a fire hose at an angle of 30.0° above the horizontal. If the speed of the stream is 40.0 m/s, at what height will the water strike the building?



We can get the time of flight from the horizontal component:

$$V\cos\theta = \frac{s}{t}$$

$$\therefore t = \frac{s}{V\cos\theta} = \frac{50}{40\cos 30} = \frac{50}{40 \times 0.866} = 1.44\ \text{s}$$

For the vertical component we can use:

$$h = ut + \frac{1}{2}at^2$$

This becomes:

$$h = 40\sin 30t - \frac{1}{2}gt^2$$

$$\therefore h = 20 \times 1.44 - \frac{1}{2} \times 9.81 \times 2.073$$

$$h = 28.8 - 10.17 = 18.63\ \text{m}$$

7. A soccer player kicks a ball horizontally off a cliff 40.0 m high into a pool of water. If the player hears the sound of the splash 3.00 s later, what was the initial speed given to the rock? Assume the speed of sound in air to be 343 m/s.

The time of flight of the rock follows from

$$y_f = y_i + v_{yi}t + \tfrac{1}{2}a_y t^2$$

$$-40.0\text{m} = 0 + 0 + \tfrac{1}{2}(-9.80\text{m/s}^2)t^2$$

$$t = 2.86\text{s}$$

The extra time $3.00\text{s} - 2.86\text{s} = 0.140\text{s}$ is the time required for the sound she hears to travel straight back to the player. It covers distance

$$(343\text{m/s})0.143\text{s} = 49.0\text{m} = \sqrt{x^2 + (40.0\text{m})^2}$$

where x represents the horizontal distance the rock travels. Solving for x gives $x = 28.3\text{m}$. Since the rock moves with constant speed in the x direction and travels horizontally during the 2.86s that it is in flight,

$$x = 28.3\text{m} = v_{xi}t + 0t^2$$

$$\therefore v_{xi} = \frac{28.3\text{m}}{2.86\text{s}} = 9.91\text{m/s}$$

8. A boy can throw a ball a maximum horizontal distance of 40.0 m on a level field. How far can he throw the same ball vertically upward? Assume that his muscles give the ball the same speed in each case.

A boy can throw ball 40 m vertically upward. The greatest horizontal distance he can throw the same is ($g=10ms^{-2}$)

Given: $H = 40m$

Now, $H = \frac{u^2}{2g}$ —(1)

The maxm. horizontal distance,

$R = \frac{u^2}{g}$ —(2)

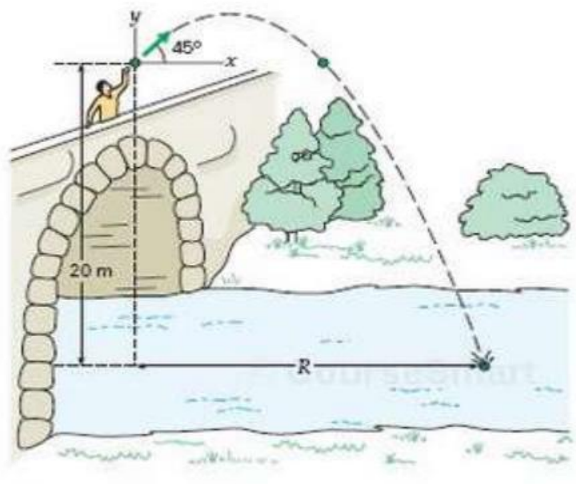
From (1) and (2), we get

$R = 2H$

$R = 2 \times 40$

$\therefore R = \underline{80m}.$

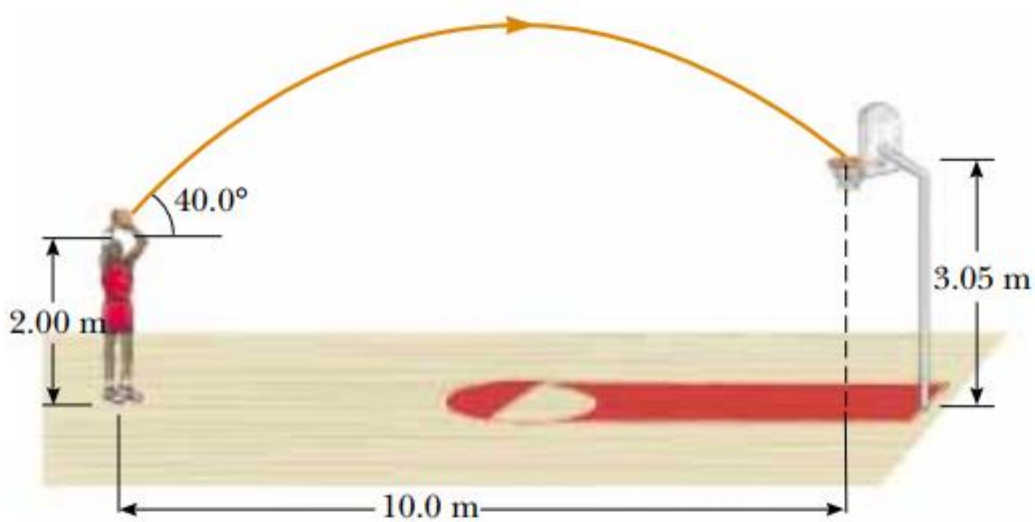
9. A stone thrown off a bridge 20 m above a river has an initial velocity of 12 m/s at an angle of 45 degrees above the horizontal. The stone lands in the river below.




(a) what is the range of the stone?

(b) at what velocity does the stone strike the water?

10. A basketball player throws a ball to score through the ring as shown in the figure below. Determine the time taken for the ball to just pass through the inside of the ring.



The components of the initial velocity are

$v_{ix} = v_i \cos 40$, and $v_{iy} = v_i \sin 40$

the time for the ball to move 10.0 m horizontally is

$$t = \frac{\Delta x}{v_{ix}} = \frac{(10.0)}{v_i \ \cos 40}$$

At this time, the vertical displacement of the ball must be

$$\Delta y = (3.05 - 2.00)\ \text{m} = 1.05\ \text{m}\ .$$

Thus, $\Delta y = v_{iy}\, t + \frac{1}{2} a_y t^2$ becomes

$$1.05\ \text{m} = v_i \sin 40\ \frac{10.0}{v_i \ \cos 40} + \tfrac{1}{2}(-9.8)\frac{(10.0)^2}{(v_i \ \cos 40)^2}$$

Which yields, $v_i = 10.7\ \text{m/s}^2$